MEDINOVA TEACHING HOSPITAL

Class 07 — Gastroenterology

GI Bleeding · Liver Disease · Acute Abdomen · Pancreatitis

> *"The abdomen conceals more than it reveals."*

CLERKSHIP SERIES | MODULE 07 of 20

SECTION 01

Acute Upper GI Bleeding (UGIB) Management

UGIB is a medical emergency requiring rapid risk stratification. The **Glasgow-Blatchford Score (GBS)** is used pre-endoscopy, while the **Rockall Score** is used post-endoscopy.

Rockall Component	0 Points	1 Point	2 Points	3 Points
Age	< 60	60 - 79	≥ 80	-
Shock	None (HR < 100)	Tachycardia (HR > 100)	Hypotension (BP < 100)	-
Comorbidity	None	-	IHD / Heart Failure	Renal/Liver Failure
Diagnosis	Mallory-Weiss	Other lesions	Malignancy	-

Variceal vs. Non-Variceal Strategy

Variceal (Portal Hypertension):
• Terlipressin (2mg IV every 4h) + Prophylactic Antibiotics (Ceftriaxone).
• Urgent Endoscopy within 6-12 hours for Band Ligation.

Non-Variceal (Peptic Ulcer):
• High-dose PPI (80mg Omeprazole bolus → 8mg/h infusion).
• Endoscopy within 24 hours for Clipping/Thermal Coagulation.

> **Prof. Obi's Pearl #1** "A drop in Haemoglobin in an acute bleed is a late sign. Treat the pulse and the blood pressure, not the lab result from four hours ago."

SECTION 02

Chronic Liver Disease & Cirrhosis

Child-Pugh Classification (Predicts 1-2 Year Survival)

Parameter	1 Point	2 Points	3 Points
Encephalopathy	None	Grade 1-2	Grade 3-4
Ascites	None	Mild/Moderate	Severe (Tense)
Bilirubin (µmol/L)	< 34	34 - 50	> 50
Albumin (g/L)	> 35	28 - 35	< 28
INR	< 1.7	1.7 - 2.3	> 2.3

Class A: 5-6 pts | Class B: 7-9 pts | Class C: 10-15 pts (Decompensated).

Complications & Management

- **SBP (Spontaneous Bacterial Peritonitis):** Infection of ascites. Neutrophils > 250/mm³. Rx: Cefotaxime.
- **HRS (Hepatorenal Syndrome):** Renal failure in liver disease. Rx: Terlipressin + Albumin.
- **HE (Hepatic Encephalopathy):** Brain dysfunction due to ammonia. Rx: Lactulose + Rifaximin.

> **Prof. Obi's Pearl #2** "The liver is a resilient organ, but once it fails, the kidneys are next. Always monitor the creatinine in a cirrhotic patient like your life depends on it."

SECTION 03

Acute Abdomen Assessment

Differentiating "Surgical" (requires theatre) from "Medical" causes is the primary goal.

Sign of Peritonism	Description	Likely Pathology
Guarding	Involuntary contraction of abdominal muscles.	Underlying inflammation.
Rigidity	"Board-like" abdomen, completely immobile.	Perforation (e.g., Perforated Peptic Ulcer).
Rebound Tenderness	Pain worse on release of pressure than application.	Parietal peritoneal irritation.

SECTION 04

Acute Pancreatitis

Diagnosis requires 2 of 3: Epigastric pain, Amylase/Lipase > 3x normal, or CT findings.

The Glasgow Score (PANCREAS) for Severity

PaO2 < 8 | **A**ge > 55 | **N**eutrophils (WBC) > 15 | **C**alcium < 2 | **R**enal (Urea) > 16 | **E**nzymes (AST/ALT) > 200 | **A**lbumin < 32 | **S**ugar > 10.

Score ≥ 3 suggests severe pancreatitis and needs HDU/ICU referral.

Management Principles:
1. Aggressive Fluid Resuscitation (Isotonic crystalloids).
2. Analgesia (IV Opioids).
3. Nutrition (Enteral feeding is preferred over TPN).

SECTION 05

Inflammatory Bowel Disease (IBD) Comparison

Feature	Crohn's Disease	Ulcerative Colitis (UC)
Location	Mouth to Anus (Skip lesions)	Rectum upwards (Continuous)
Depth	Transmural (Full thickness)	Mucosal only
Histology	Non-caseating Granulomas	Crypt Abscesses
Complications	Fistulae, Strictures, B12 deficiency	Toxic Megacolon, Malignancy risk

SECTION 06

Jaundice & LFT Patterns

Type	Bilirubin	ALT / AST (Hepatocellular)	ALP / GGT (Cholestatic)
Pre-Hepatic (Haemolysis)	Unconjugated ↑	Normal	Normal
Hepatic (Hepatitis)	Mixed	Marked ↑↑↑	Mild ↑
Post-Hepatic (Obstruction)	Conjugated ↑	Mild ↑	Marked ↑↑↑

> **Prof. Obi's Pearl #3** "Painless jaundice in an elderly patient is cancer of the head of the pancreas until proven otherwise. Do not wait for pain to appear; order the CT scan."

SECTION 07

Gastrointestinal Emergencies

1. Bowel Obstruction:
- Sxs: Vomiting, Absolute constipation, Distension.
- Dx: AXR (Small bowel > 3cm, Large > 6cm).
- Rx: "Drip and Suck" (IV Fluids + NG Tube).

2. Mesenteric Ischaemia:
- Sxs: "Pain out of proportion to findings".
- Dx: CT Angiography.
- Rx: Emergency Laparotomy + Resection.

3. Perforation (Emergency):
Sudden onset severe pain. Look for *Rigidity* and *Air under the diaphragm* on erect CXR. Rx: NPO, IV Antibiotics, Surgical consult.

> **Prof. Obi's Pearl #4** "The nasogastric tube is both a diagnostic and a therapeutic tool. If you suspect obstruction and you haven't put one in, you haven't finished the exam."

Medical Definitions:
- **Terlipressin:** A hormone that constricts splanchnic vessels to stop variceal bleeding.
- **Skip Lesions:** Healthy segments of bowel between diseased segments (seen in Crohn's).
- **MELD Score:** Model for End-Stage Liver Disease (used to prioritize transplant).
- **ERCP:** A procedure to remove gallstones from the bile duct via endoscopy.

OFFICIAL SIGN-OFF

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